

Junaid Ahmed Mohammed

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with 5+ years of enterprise experience building scalable Azure, Snowflake, and Databricks data platforms, complemented by graduate research in streaming systems and AI-enabled data pipelines. Designed and optimized high-throughput ETL/ELT workflows and modern lakehouse architectures processing 12 TB+ of retail data across 625+ stores in 60+ countries, delivering 40% pipeline runtime reduction and 25% infrastructure cost savings. Strong expertise in PySpark, Delta Lake, Kafka, Airflow, dimensional modeling, and multi-source enterprise data integration across batch and near-real-time workloads. Experienced in enabling AI/ML systems through vector search pipelines, LLM data preparation, and distributed data processing frameworks.

CORE COMPETENCIES & TECHNICAL STACK

Data Engineering: ETL/ELT, Data Warehousing, Lakehouse Architecture, Dimensional Modeling, SCD Type 1/2, Data Governance, Data Quality, Performance Optimization, CI/CD, Unit/Regression Testing, Source-to-Target Reconciliation, DataOps

Cloud & Platforms: Azure Data Factory, Azure Databricks, ADLS Gen2, Azure Blob Storage, Snowflake (Zero-Copy Cloning, Time Travel), Azure DevOps, Azure Key Vault, Azure ML

Data Processing: Apache Spark, PySpark, Spark SQL, Delta Lake, Apache Kafka, Apache Airflow, Scala

Programming & SQL: Python, SQL, T-SQL, PL/SQL, Stored Procedures, Power BI, Scala

Databases & Storage: SQL Server, Oracle, PostgreSQL, Teradata

AI/ML Enablement: Vector Search, pgvector, OpenAI Embeddings API, LLM Fine-Tuning (QLoRA), Azure ML

Tools & Collaboration: Git, GitHub, Confluence, Azure DevOps

PROFESSIONAL EXPERIENCE

NISUM TECHNOLOGIES, INC.

Nov 2018 – Dec 2023

Senior Data Engineer

Hyderabad, India

Led data engineering across 5 concurrent enterprise programs for Williams-Sonoma, Inc. (625+ stores, 60+ countries) and Macy's. Acted as primary technical decision-maker on pipeline architecture, data modeling standards, and Snowflake platform design, collaborating directly with WSI engineering managers and business stakeholders on requirements, timelines, and delivery.

- **Designed** scalable Azure-Snowflake lakehouse architectures supporting enterprise retail analytics, supply chain operations, customer delivery forecasting, and product management workloads across global business units spanning 625+ stores and 60+ countries.
- **Reduced** Databricks pipeline runtime by **40%** through PySpark optimization, partition pruning, broadcast joins, and Spark SQL tuning across 160+ ADF pipelines spanning a Bronze–Silver–Gold Delta Lake medallion architecture.
- **Co-architected** a Snowflake enterprise data warehouse and lakehouse consolidating **12 TB+** from Oracle, Teradata, Kafka, NetSuite ERP, and SFTP into ADLS Gen2, supporting daily batch and near-real-time ingestion across 6+ enterprise source systems.
- **Established** dimensional models, SCD Type 2 pipelines, clustering keys, and right-sized Snowflake virtual warehouses across the lakehouse, **cutting infrastructure costs by 25%**.
- **Owned the ECDD platform** (customer-facing delivery date model for Williams-Sonoma's global B2B and retail operations) — modeled carrier and routing dimensions with SCD Type 2 and applied SQL date calculation and exception-handling logic, **reducing delivery inaccuracies by 30%** and improving on-time fulfillment across 60+ countries.
- **Led** Oracle-to-Snowflake WMOS migration across warehouse, DBA, and data science teams; executed ADF orchestration, Databricks staging pipelines, Scala Snowflake loaders, and T-SQL stored procedures for metric computation.
- **Architected** a centralized PIM platform as single source of truth for enterprise product data, integrating Sterling OracleDB, SAP, and SFTP feeds across Orders, Products, Stores, and Vendor domains; drove cross-functional alignment between supply chain, merchandising, and engineering on data model requirements, delivering structured datasets to Power BI and retail analytics reporting layers.

- **Standardized** data quality, governance (SCD deduplication), and Git-based CI/CD deployments via Azure DevOps; enforced daily batch SLA compliance and secured pipeline credentials via Azure Key Vault.
- **Served as technical lead for 7 junior engineers** — conducted code reviews, defined pipeline engineering standards, and provided architecture guidance; coordinated feature dataset delivery into Azure ML workflows and maintained **100% on-time sprint delivery** across 5 concurrent projects and quarterly Agile releases.

Client: Macy’s, Inc. | *Early-career data engineer supporting enterprise retail data warehouse.*

- **Monitored** 25+ ADF pipelines across daily batch windows; executed source-to-target reconciliation to detect data loss and duplication, debugged pipeline failures, and coordinated production fixes with senior engineers to maintain SLA compliance.
- **Performed** unit and regression testing on pipeline transformations prior to production deployments; profiled 20+ operational tables in Oracle and SQL Server to certify data readiness and resolved historical reload issues.

INDIANA UNIVERSITY INDIANAPOLIS

Jan 2024 – Dec 2025

Graduate Research Assistant, COMET Lab

Indianapolis, IN

Owned the data engineering workstream for applied ML and public health research, designing and operating scalable data pipelines and orchestration infrastructure independently.

- **Architected** an end-to-end streaming ingestion platform using Kafka and PySpark Structured Streaming, validating **50K+ rows per batch cycle** against revenue, cost, date, and categorical business rules before loading into PostgreSQL.
- **Orchestrated** automated model training via Airflow DAG with threshold-triggered deduplication and distributed Spark-submit jobs for revenue and units-sold forecasting; containerized the full stack with Docker Compose.
- **Constructed** LLM training data pipelines and evaluation infrastructure for QLoRA fine-tuning of LLaMA 3.2 and Qwen 2.5; built structured prompt datasets and systematized correctness harnesses to validate domain-specific Python code generation.
- **Delivered** multi-source data ingestion pipelines for NHANES and IPUMS public health datasets; standardized file formats using PySpark, enforced weighted aggregation logic, and streamlined report-ready data delivery for research and policy audiences.

PERSONAL PROJECTS

JOB AI MATCHER | *AI-Powered Semantic Job Intelligence Platform*

In Development | Beta

Stack: Python, FastAPI, PostgreSQL, pgvector, OpenAI Embeddings API

- **Built and Deployed** a cloud-hosted semantic job matching platform using FastAPI and PostgreSQL; engineered a vector search layer with pgvector and OpenAI embeddings to rank resumes against live job descriptions via similarity search.
- **Integrated** live job ingestion from multiple external APIs with an end-to-end pipeline covering semantic matching, ranking, and tailored cover letter generation via GPT-4o-mini.

EDUCATION

INDIANA UNIVERSITY

Indianapolis, Indiana

M.S. in Applied Data Science | GPA: 3.97 / 4.0

Aug 2023 – Dec 2025

Coursework: Statistical Learning, Cloud Computing, NLP, Computer Vision, Database Design

CERTIFICATIONS

Snowflake SnowPro Core | Azure Data Engineer Associate (DP-203) | Generative AI 101, Indiana University